Algebra and Discrete Mathematics (ADM)

Tutorial 8 Eulerian tours

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Graph

Definition

A *graph* consists of two sets: $V(G)$, called the *vertex set*, and $E(G)$, called the edge set. An *edge*, denoted xy, is an unordered pair of vertices $x, y \in V(G)$.

Theorem (Handshaking Lemma)

Let $G = (V, E)$ be a graph. Suppose $V = \{v_1, v_2, \ldots, v_n\}$, then

$$\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2|E|$$

Draw a graph

Question

Draw a graph with 6 vertices such that their degrees are

$$1,\ 2,\ 3,\ 4,\ 5,\ 2$$

We have

$$\sum_{v} \deg(v) = 1 + 2 + 3 + 4 + 5 + 2 = 17,$$

which is odd. Hence, such a graph does not exist.

Draw a graph

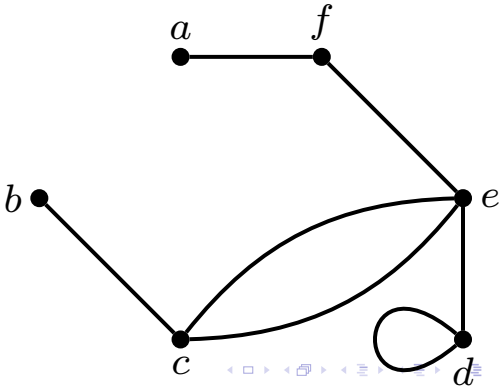
Question

Draw a graph with 6 vertices such that their degrees are

$$1,\ 1,\ 3,\ 3,\ 4,\ 2$$

$\deg(a) = 1, \quad \deg(b) = 1, \quad \deg(c) = 3,$

$\deg(d) = 3, \quad \deg(e) = 4, \quad \deg(b) = 2,$



Draw a graph

Question

Construct a graph to represent a club with 13 members, where each member is acquainted with exactly 5 other members.

The sum of degrees is given by

$$13 \times 5 = 65$$

which is odd. Hence, such a graph does not exist.

Non-existence of certain regular graphs

Statement

Prove that a k-regular graph, where k is odd, with an odd number of vertices does not exist.

Proof.

If a graph has n vertices and is k-regular, then the sum of all vertex degrees is

$$nk.$$

Since both n and k are odd, the product nk is odd.
But by the Handshaking Lemma, the sum of the degrees of all vertices must be even.
This is a contradiction, so such a graph cannot exist. □

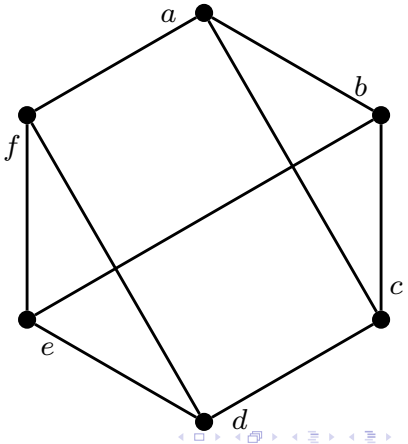
Draw a graph

Question

Draw a graph with 6 vertices, each of degree 3

$$\frac{6 \times 3}{2} = 9$$

so the graph must have 9 edges.



Draw a graph

Question

Draw a graph with 7 vertices, each of degree 3

$$7 \times 3 = 21,$$

which is odd. Hence, such a graph does not exist.

Draw a graph

Question

Draw a graph with 4 vertices, each of degree 1

Note that this graph is not connected.

Draw a graph

Question

Draw a graph with 6 edges, 4 vertices with degrees

$$1,\ 2,\ 3,\ 4$$

It must hold that

$$2|E| = \sum_{v \in V(G)} \deg(v)$$

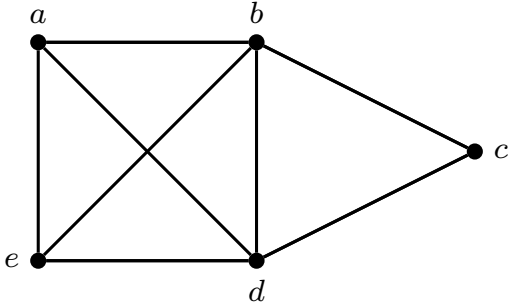
However

$$2|E| = 2 \times 6 = 12$$

$$\sum_{v \in V(G)} \deg(v) = 1 + 2 + 3 + 4 = 10$$

Hence, such a graph does not exist.

Touring a graph



- Path: does not repeat vertices
  - Example: $a,\ b,\ c,\ d,\ e$
- Trail: does not repeat edges
  - Example: $a,\ b,\ e,\ a,\ d,\ c$
- Cycle: a closed path
  - Example: $a,\ b,\ c,\ d,\ e,\ a$
- Circuit: a closed trail
  - Example: $a,\ b,\ c,\ d,\ b,\ e,\ a$

Eulerian and semi-Eulerian

Theorem

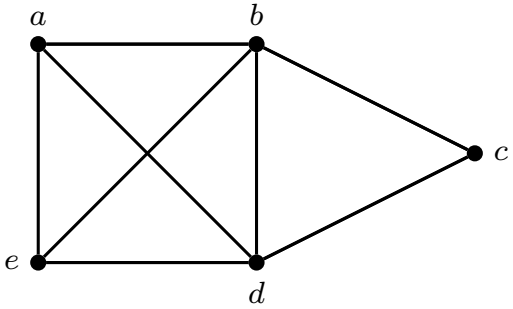
A graph G is Eulerian if and only if

- *G is connected and*
- *every vertex has an even degree*

A graph G is semi-Eulerian if and only if

- *G is connected and*
- *exactly two vertices have odd degree*

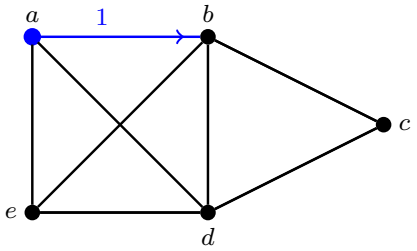
Semi-Eulerian graph

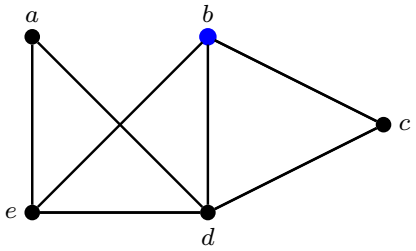


$$\deg(a) = 3, \quad \deg(b) = 4, \quad \deg(c) = 2, \quad \deg(d) = 4, \quad \deg(e) = 3$$

The graph is semi-Eulerian

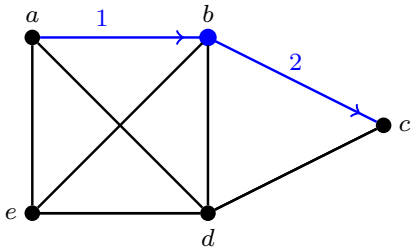
Fleury's Algorithm

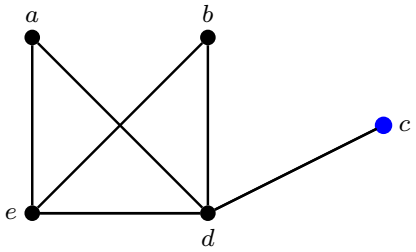




- Step 1. Start at vertex a
- Step 2. Choose edge ab
- Step 3. Travel to vertex b and delete edge ab. Current vertex: b.

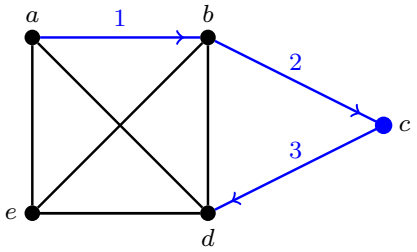
Fleury's Algorithm

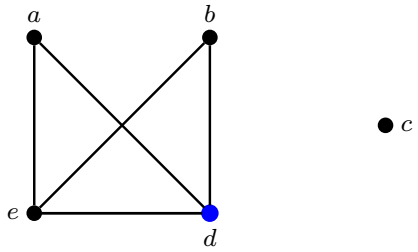




- Step 2. Choose edge bc
- Step 3. Travel to vertex c and delete edge bc. Current vertex: c.

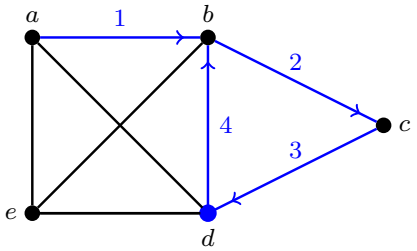
Fleury's Algorithm

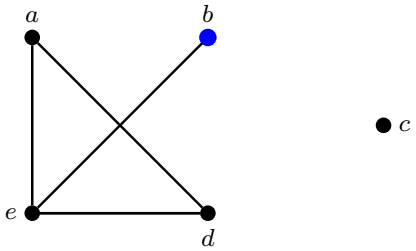




- Step 2. The only available choice is edge cd.
- Step 3. Travel to vertex d and delete edge cd. Current vertex: d.

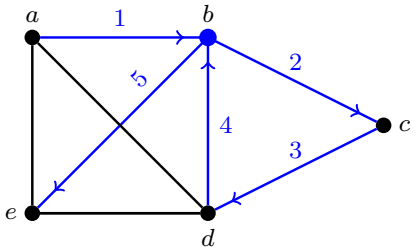
Fleury's Algorithm

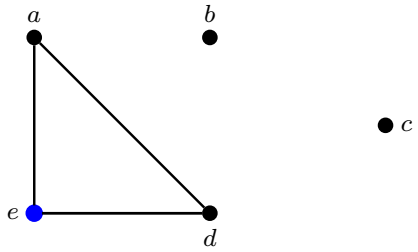




- Step 2. Choose edge db
- Step 3. Travel to vertex b and delete edge db. Current vertex: b.

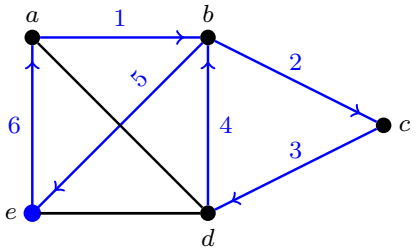
Fleury's Algorithm

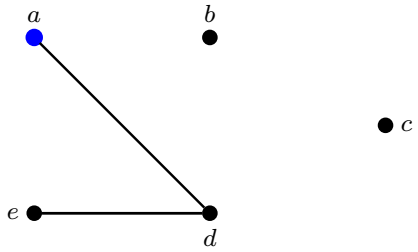




- Step 2. The only available choice is be
- Step 3. Travel to vertex e and delete edge be. Current vertex: e.

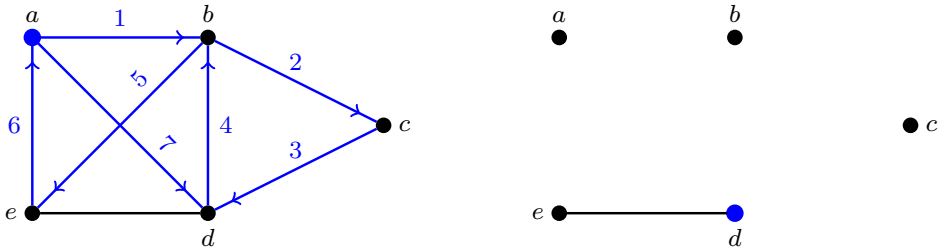
Fleury's Algorithm





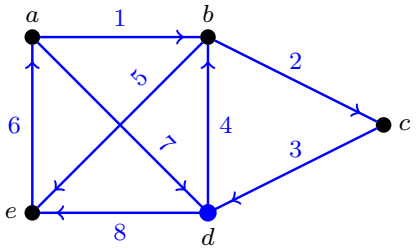
- Step 2. Choose ea
- Step 3. Travel to vertex a and delete edge ea. Current vertex: a.

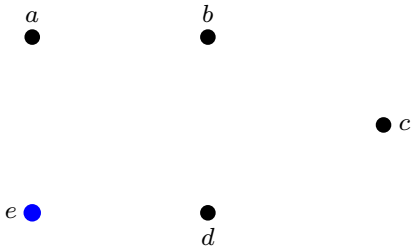
Fleury's Algorithm



- Step 2. The only available choice is ad
- Step 3. Travel to vertex d and delete edge ad. Current vertex: d.

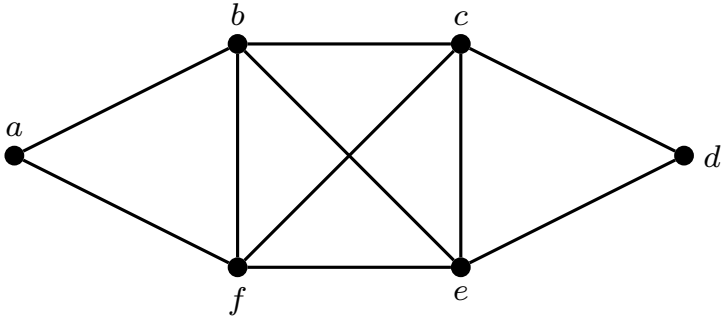
Fleury's Algorithm





- Step 2. The only available choice is de
- Step 3. Travel to vertex e and delete edge de.
- We have obtained an Eulerian trail

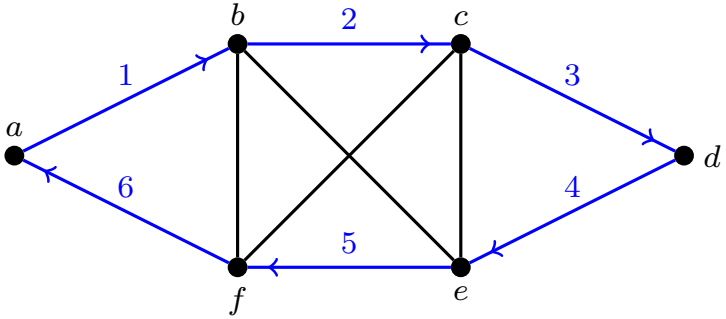
Hierholzer's Algorithm



$$\deg(a) = 2, \quad \deg(b) = 4, \quad \deg(c) = 4, \quad \deg(d) = 2, \quad \deg(e) = 4, \quad \deg(f) = 4$$

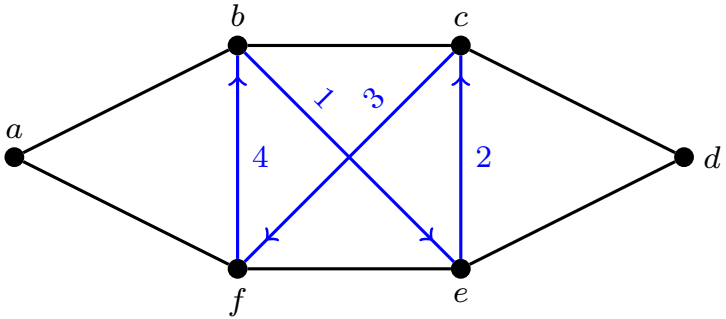
The graph is Eulerian.

Hierholzer's Algorithm



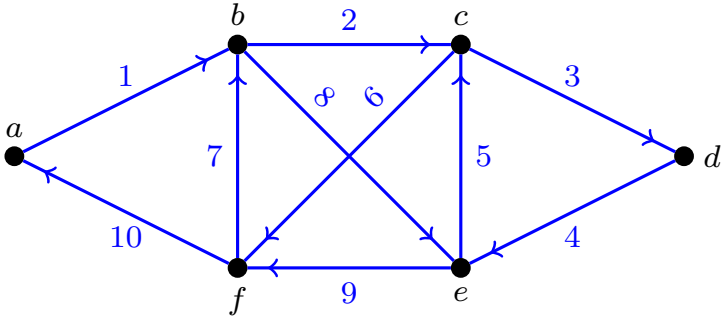
- Step 1. Choose vertex a and find a circuit starting at a.

Hierholzer's Algorithm



- Step 2. Since $\deg(b) = 4$, two unused edges remain incident to b, so we can find a second circuit starting at b.

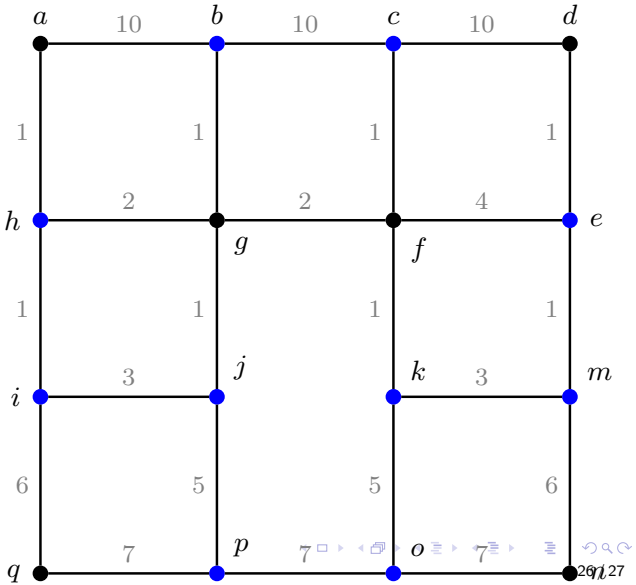
Hierholzer's Algorithm

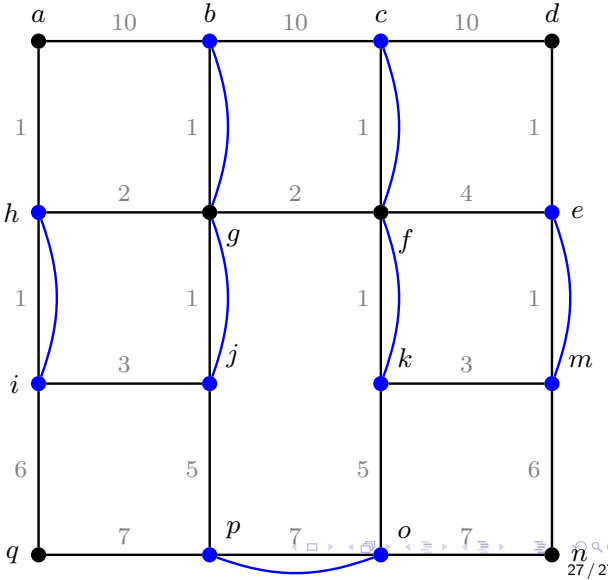


- Step 3. Combine the new circuit with the existing one.
- We have obtained an Eulerian circuit.

Chinese Postman Problem

- Find an optimal Eulerization, where the edge weights represent cost.
- The unweighted version suggests first pairing odd vertices using as few duplicated edges as possible.
- We first attempt to pair vertices along paths using edges of weight 1 as much as possible
- We are able to do this for all vertices except o and p, and we duplicate edge op



Chinese Postman Problem

- First, pair odd vertices using paths of weight 1 whenever possible.
- This works for all odd vertices except o and p, so we duplicate edge op.
- If we avoid duplicating op, then o and p must be paired through edges such as ko and jp, each of weight 5.
- Therefore, no Eulerization has total added weight less than 13.